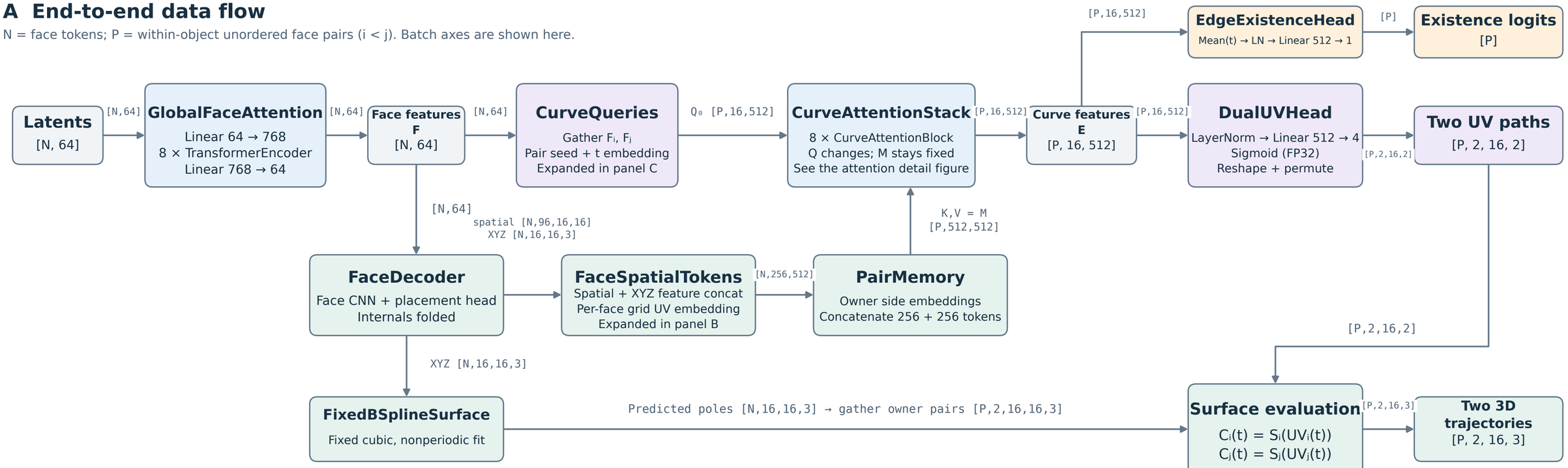
Current curve decoder: shared queries, two UV trajectories

Source: src/brepnet/models/vae/uv_intersection.py | Current 64-D configuration | No proposed modules

A End-to-end data flow

N = face tokens; P = within-object unordered face pairs (i < j). Batch axes are shown here.



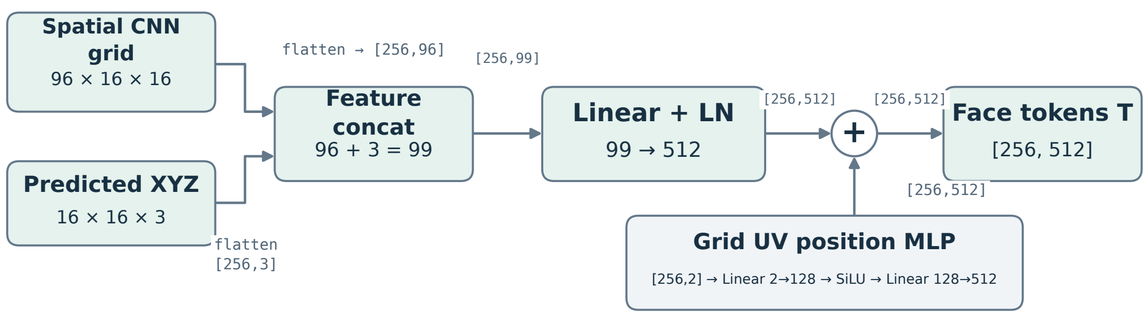
Global attention: d = 768, 16 heads, FFN = 2048, pre-LN; attention is masked across objects.

Not one guaranteed shared 3D curve

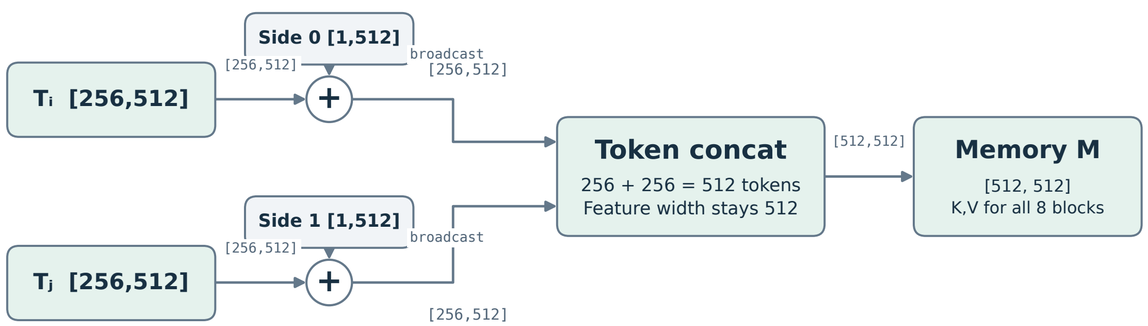
B Memory construction

Per-face / per-pair views below omit the N or P batch axis. ⊕ means elementwise addition.

B1 FaceSpatialTokens — concatenate features, then add grid position

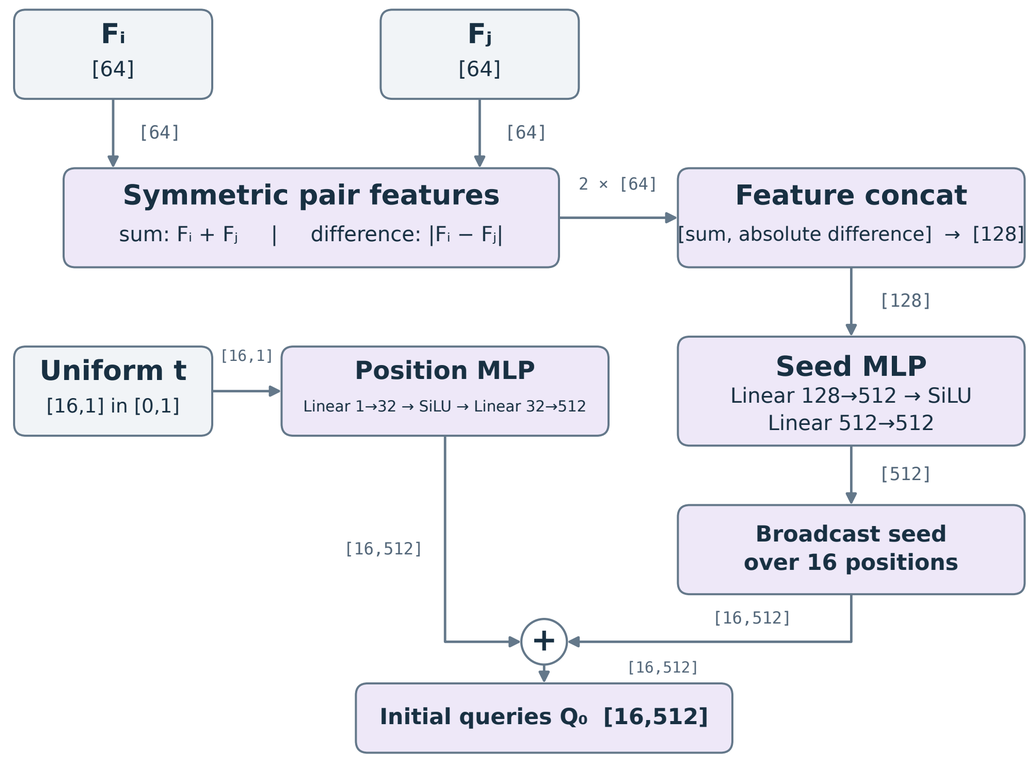


B2 PairMemory — add owner identity, then concatenate tokens



C Query construction

One shared 16-position query set per unordered pair. These are features, not geometric curve points.



All three concatenations are explicit: (1) 96+3 features, (2) 256+256 tokens, (3) 64+64 pair features. Query and memory are never concatenated.